

AWS ECS (Elastic Container Service)

01-ml-infrastructure/ecs

Overview

ECS is AWS managed container orchestration. It schedules containers onto compute capacity and keeps services running according to the task definition you declare.

For ML applications, ECS is a practical choice for serving models, running supporting APIs, and executing recurring preprocessing or postprocessing jobs without managing Kubernetes.

Core Concepts

- Core objects include clusters, task definitions, tasks, and services. A task definition describes the containers, CPU, memory, networking, secrets, and startup behavior.
- ECS can run on EC2 instances that you manage or on Fargate, where AWS manages the underlying servers.
- Services maintain a desired number of healthy tasks and integrate with load balancers, service discovery, and deployment strategies.

How It Works

- You package the application as a container, store it in ECR, create a task definition, and then run it as a task or a service.
- For a long running inference API, ECS typically deploys tasks behind an Application Load Balancer and scales them based on CPU, memory, request count, or custom metrics.
- For scheduled or event driven jobs, ECS can launch short lived tasks that process a queue, transform data, or prepare artifacts for later training or inference stages.

AI and ML Relevance

- ECS works well for FastAPI model servers, embedding services, feature preprocessing workers, and internal tools that need container consistency but not the full weight of a larger platform.
- If your inference stack depends on custom native libraries or multiple sidecars, ECS often gives a cleaner operational model than forcing everything into a serverless boundary.

Operational Notes

- Right size task resources and container concurrency. Overcommitting CPU or memory leads to noisy failures and bad latency under load.
- Use IAM task roles, central logging, health checks, and rolling deployments so the service can update without downtime.
- Fargate reduces operational work, but EC2 backed ECS may be cheaper or more flexible for heavy GPU or host level customization requirements.

What To Remember

- Task definition is the runtime contract. Service is the controller that keeps tasks alive.
- Fargate optimizes convenience; EC2 launch type optimizes control.
- ECS is a strong middle ground when you want containers without adopting Kubernetes complexity.